Oculus Tray Tool
Version 0.87.7

User Guide

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Contents

Introduction

I made this tool to help people get the most out of their Oculus Rift headset, and to get the best possible experience out of this amazing piece of hardware. The Oculus Tray Tool is not an official Oculus software. This tool is free and always will be.

Special Thanks

I want to thank user headkaze over at http://headsoft.com.au for suppling the icons, the setup script and providing some general programming advice. I also want to thank Mike from Virtual Reality Oasis and F4CEpa1m for their videos on Oculus Tray Tool. Check out their Youtube channels for more awesome VR stuff.

Virtual Reality Oasis channel: https://www.youtube.com/channel/UCsmk8NDVMct75j_Bfb9Ah7w
F4CEpa1m's channel: https://www.youtube.com/channel/UCv6ftEMDFPpzwf4TjkIfiJA

How It Works

Oculus Tray Tool relies on an Oculus supplied application called Oculus Debug Tool. With this application developers and others can set a higher Pixel Density (Super Sampling) as well as other options. Oculus Tray Tool uses this application in the background to work it's magic.
Changing Pixel Density is done on the Oculus Service itself, not the game in question. Any VR app that is started *after* this has been done will get the current Pixel Density setting.

With the Profiles, OTT monitors the execution of all processes on the computer and if any process match a profile it will apply those setting to the Oculus Service, and the game will then "inherit" those settings. Most games take a little while to startup and OTT can usually catch this before the game has fully loaded and all will work fine. As OTT needs to catch the game fast, before it has fully loaded, the monitoring needs to happen at a very fast rate, causing a bit of CPU usage.
When a profile has been detected the monitoring process will stop and CPU usage should drop.

In some cases, if a game or app starts too fast, or simply does not support changing Pixel Density it will not work. In cases where you notice a Profile for a game does not get applied, try changing the Pixel Density on the main GUI before starting the game, or switch Detection Method on the Profile.

You can also use the Voice Commands to change Pixel density / Super sampling before starting the game, so you don't even have to take off your headset. The Oculus Debug Tool is shipped with the Oculus Tray Tool installation, but will by default use the one that is shipped with Oculus as that will always be the newest. If for some reason there is an issue, there are options to use the one shipped with OTT instead.

Installing Oculus Tray Tool

Installation is done by running the OTTSetup.exe installer.

Updating from a previous release

Simply run the new installer and it will remove the previous version (see Uninstalling below)
And install the new version. When the new version starts up for the first time it will migrate
All settings to the new version, provided you did not remove them as part of the uninstall.

Uninstalling

To uninstall, first uncheck the "Start with Windows" checkbox, if you have that set.
Then uninstall in the way you do any applications. Pay attention to the dialog boxes that
ask if you want to remove all settings and databases. The settings contain all the different

choices you have made in OTT, like what boxes are ticked and so on. The Database contains all your Profiles. If you are upgrading to a new release you might want to preserve both the Settings and the Databases.

Automatic Update Checking

OTT automatically checks for updates each time it starts. If an update is found a popup will notify you. Click the popup toast or check the Updates tab to see the update and download the new version. You can opt-out of automatic update checking by unticking the "Check for updates on startup" checkbox on the Tray Tool tab. This restricts update checking of the main application, as well as the Icons and Steam game databases, to manual mode only. You can manually check for updates using the "Check for updates" button on the Advanced tab.

Donate

If you want to make a donation to this project you can use the PayPal button on the main GUI.

Settings

Oculus Tray Tool has many settings, but they are not very complicated. The tool is designed to have the user, You, make the settings once and then basically forget about them.

Tray Tool Tab

- Start with Windows: Check this box to have the application start when Windows starts.
  This setting requires you to run the application as Administrator. This will add a scheduled task that runs OTT at logon. The task will run as administrator and suppress UAC prompts.
  If for some reason you cannot get OTT to start properly, and you have it set to Start with Windows and want to change that, go to Task Scheduler and find the Oculus Tray Tool task and disable it.

- Start Minimized: When checked, the application will be minimized to the system tray on startup.

- Hide from Alt + Tab: When checked, Oculus Tray Tool will not show up in the open applications list when you cycle though open applications using Alt + Tab.

- Use AudioSwitcher: This lets OTT manage your audio, mic and comunication devices. Use this to have OTT switch set the default devices to the selections made, and also switch them back to devices of your choice. Click the Configure button to set up your devices.

  You can have this switch occur when either of the below is true.

    - Oculus Tray Tool starts
    - Oculus Home starts
    - A profile loads

- Reset AudioSwitcher
  If you experience any issues with the AudioSwitcher you might need to reset it's settings. Click the "Reset" button to clear all settings and load all audio and mic devices again. Resetting the AudioSwitcher might be needed when a new version of OTT has been release that contains fixes or changes to this feature.

- Use HotKeys: With Hotkeys you can set up keys to use for showing various information panels in the headset, enable and disable ASW and to close the currently running VR application. This "registers" those keys with Windows, so whenever OTT is running it will respond to these key presses. Other applications using the same key combo might take over if started after OTT.

Power Options

- Set Power Plan on Start: Here you can set the Power Plan you want your computer to use when Oculus Tray Tool or Oculus Home starts. The consensus seems to be to use a higher performance power plan when using the Rift to get the most juice out of your system.

  NOTE: A "high" power plan disables some device from going to sleep, which might include the Rift controllers. This might cause battery power drain on the Rift controllers

- Set Power Plan on Exit: Similar to the above, this option lets you specify which Power Plan should be set when Oculus Tray Tool/Oculus Home closes.

- USB Selective Suspend: This is tied to the currently set power plan, e.g. what you have set in 'Set Power Plan on Start', and might change depending on what power plan you select. Once OTT changes the powerplan, it will check the USB Selective Suspend setting for that plan and display the current state in this drop-down list. You want to set this to 'Disabled' to prevent Windows from cutting power to your USB devices, which can cause the Rift to disconnect. Once you have set a USB Selective Suspend value for a specific powerplan it will not change, unless you do that manually.

  NOTE: This might cause battery power drain on the Rift controllers

- Rift Power Management
  You will get warnings in the log window if any of the following devices have power management enabled.

    - Rift
    - Sensors
    - Xbox controller

  You can right-click on any of these messages in the log window and select "Disable Power Management". This will disable PM on all the reported devices.
  If you want to disable PM on any other USB devices, see below for instruction.
  You can also select to always disable PM on sensors, as PM might be re-enabled

when Oculus is updated.

NOTE: This might cause battery power drain on the Rift controllers

- Disabling Power Management on USB HUBs

    - Right-click ”My computer/This PC” and select Manage.
    - In the left pane, select Device Manager.
    - Scroll to the bottom of the right pane and find “Universal Serial Bus controllers” and expand it by clicking the little arrow on the left.
    - Go through all of the listed devices, right-click on them and select “Properties”.
    - Select the Power Management tab if there is one. If not, just close the window and proceed to the next device in the list.
    - Uncheck the “Allow the computer to turn off this device to save power” checkbox.

- Enable Ultimate Performance Power Plan
  Follow these steps to enable Ultimate Performance Power Plan in Windows 10. This mode will make applications perform a little faster. It essentially disables every single power-saving feature imaginable, letting you enjoy a slight performance boost.

  1) Make sure you have updated your PC to Windows 10 version 1803. You can check this in Settings > System > About.

  2) Open Settings > System > Power & sleep > Additional Power Settings

  3) Under Choose to customize a power plan, expand the option which says “Unhide Additional plans.”

  If you do not see Ultimate Performance Mode, then follow the next steps.

  1) Open a Command Prompt as administrator.

  2) In the command prompt, copy the following command, and hit Enter.

  ```
  powercfg -duplicatescheme e9a42b02-d5df-448d-aa00-03f14749eb61
  ```

  2) You should now be able to select Ultimate Performance as power plan.

Game Settings

- Profiles: Click the View & Edit button to open the Profiles window.
  Here you should see your Profiles once you have created them. Create a new Profile by clicking Create New Profile or right-clicking on an empty space in the list and selecting the appropriate choice. Select a game from the drop-down list and then choose your options. You can also select “’- All Games & Apps -“ to create profiles for all available games in one go. These profiles will all be created with the same settings, that you can later change to the desired values.
  If the game is not listed here it may be a third-party app that was not installed using Oculus Home or Steam, and in such cases Oculus Tray Tool cannot find it. Use the three-dots Browse button to find the main executable for the game. You can also change the name of the app as it appears in the Profiles list. The name is by default the same as the executable you select.

You can edit settings for multiple profiles or delete multiple profiles at once by checking the box to the left of the profile, and then right-click to bring up the menu and select the appropriate choice for your action.

- Super Sampling: Select a Super Sampling value in the dropdown. This value will be applied when the selected game/application is detected as running.
- ASW Mode: Select what Mode Asynchronous Spacewarp should use for this profile.
    - Set Delay
      This is the number of seconds to wait before applying ASW after detecting the game as running. Some games do not like when you change ASW too fast so in some cases increasing this value is needed.
- CPU Priority: Set this to a higher value than Normal to give this app additional CPU priority. This can help with performance of the game. A setting of "Default" will leave the CPU priority as it was when the game was started. Any other setting will apply, even if the game is already running with that Priority.
    - Set Delay
      This is the number of seconds to wait before changing CPU Priority after detecting the game as running. Some games do not like when you change CPU Prio too fast so in some cases increasing this value is needed.
- Detection method: Sets the method to use when detecting application starts.
  WMI is the default one. This consumes less CPU, but is less accurate.
  Timer consumes more CPU but is also more accurate. Use WMI to start with, and if you are having issues with getting the profiles for a particular app to apply, try switching to Timer and see if that works better. Or, launch the game directly from the Oculus Tray Tool Library view.
- Adaptive GPU Scaling: This is a technology that scales resolution in order to keep a steady framerate. Some games seem to work better with this set to Off.
- Screen Mirror:
    - Default: Do nothing and let the game show the desktop mirror as it does.
    - Minimize: Minimizes the desktop mirror of the game to the task bar.
      It might take a few seconds for the app to minimize.
    - Forced: If a game does not natively mirror to the desktop this option will use the OculusMirror app to mirror the game. The FoV Multiplier will not affect the mirror if set to Forced.
- Audio confirmation when profile is applied: Ticking this box will make Oculus Tray Tool notify you using audio when a profiled game launch is detected and the profile is applied. You will hear a voice saying "Game launch detected" when this happens. The Rift must be set as the Default audio device for you to hear the sound through the Rift headphones.

- Default Super Sampling: This value sets the Super Sampling to be applied when Oculus Home starts. This value will also carry over to any game/application that you do not have a profile for. You can also use this to change SS while in a game. Note that you will need to restart the game for the new value to be applied. Setting super sampling with this tool should override any setting set in-game. If a game has the option to set SS, OTT should override that value. In case you want to use the built-in SS you need to either set Super Sampling on the main GUI to 0 or set it to 0 in the games profile. I write “should” as in some cases, depending on how the developer of the game has done things, Super Sampling values acts as a multiplier. This means that if the game run natively at 1.0x then all is good, but if it runs at for example 1.2, and you choose a new value of 1.5, you will in reality get 1.5*1.2 = 1.8.

- Default ASW Mode: Allows you to set Asynchronous Spacewarp to Off, Auto or 45 fps with ASW On. The tool starts with ASW set to Auto. Changing this when in Home or in a VR application will take effect immediately. Use the ASW Status Overlay to confirm. If you are currently running a game that has a profile the ASW setting will be changed to that profile.

- Adaptive GPU Scaling: This is a technology that scales resolution in order to keep a steady framerate. Some games seem to work better with this set to Off. The Global setting might Not always carry over to the started game, as such this setting is also available in the Profiles.

- FOV Multiplier: Sets the FOV of the headset. Setting this to a value *higher* than 1.0 will only affect the Mirror view that is shown on your regular display. Useful when streaming gameplay and you need to make sure that the Mirror view matches what you actually see in the headset. However, setting this to a value *lower* than 1, like 0.8 for example, will actually cause a lower FOV in the headset. This will increase FPS as less pixels are being drawn, at the expense of a lower FOV in the headset. **NOTE:** This currently does **NOT** work with the **Quest** as Link needs to be restarted, making it a bit useless in Profiles. For the Quest I recommend instead using the FoV Multiplier setting on the Game Settings tab. As that is applied when OTT starts it should work fine for the Quest as OTT should be started before anything else.

- Oculus Homeless: This is a very lightweight Home replacement made by EmuVR developer and Reddit user NeoZeroo. Instead of loading the Home environment (with the Pirate ship) you will simply have solid color background. You can edit the color of the background, and also add some background music of your choice. Ticking the box for “Automatically patch” will make OTT automatically re-apply Oculus Homeless when the Oculus App is updated. Oculus Homeless uses much less CPU/GPU resources than the default Home environment.
  When you set Oculus Homeless to enabled, the original ‘Home2-Win64-Shipping.exe’ binary is backed up to the OTT installation folder, and also renamed to ‘Home2-Win64-Shipping.exe_ORG’ and kept in the original path (..\Oculus\Support\oculus-worlds\Home2\Binaries\Win64). When you disable Homeless, OTT will first try to rename the _ORG version back to it’s original name, but if that for some reason fails it will copy the backed up file to the original path. The Oculus Service will be restarted when you Enable/Disable this feature.

  NOTE: Oculus Tray Tool uses a hashing method for determining when it needs to re-apply Homeless. This hashing is done on the original ‘Home2-Win64-Shipping.exe’ binary. This hash is also calculated when Homeless is first installed. It is therefore very important that you have not manually already replaced this with a custom .exe, or OTT will calculate hashes and backup this binary instead of the original.

- Mirror Oculus Home: Setting this to Enabled will mirror the Home environment and Oculus Library to the desktop. Perfect for demoing where you need to tell your friends where to point and click. When a profile is detected the Home mirror will close and start again when the app closes.

- OVR Server Priority: This will, like the option for Profiles, increase the CPU priority of the OVRServer_x64.exe process which can increase performance in games.
  Changing OVR Server Priority also changes the priority of Oculus Dash which will run with the same priority.

- Voice commands: Set to Enabled to enable voice commands. In the current version the voice commands do not need any user training. Support for Voice Commands are done using the Microsoft Speech Platform and it is installed as part of the Oculus Tray Tool setup.

  The setup also installs the default English (US) language pack for voice recognition, but you can install other languages if you want to talk to OTT with your native tongue. If you wish to do so you also need to add spoken phrases to the Voice Commands in this language. To install additional languages, visit Microsoft via the below link.

  https://www.microsoft.com/en-us/download/details.aspx?id=27224
  Click "Download" and you will be able to select specific languages to download on the next page.

  NOTE: Because of the way the Microsoft Speech API works, the display language in Windows (the language of the menus and such) must match the installed Speech Language Pack or voice commands will not work. If you for example have English set as display language, and install a German Speech Language Pack, things will not work. In this example, you would need Either set German as display language in Windows, or use the English-US language pack.

  With voice commands you can control the following:

  - Pixel Density (Super Sampling) - Restart current game for new value to apply.
  - Asynchronous Spacewarp (ASW) - Applied immediately.
  - Visual HUD overlays - Applied immediately.
  - Launch SteamVR – Speak the phrase for "Launch SteamVR" to start SteamVR. Once launched, press either the menu button on the XBOX controller or the menu button on the Left Touch controller to access the Steam Library. When exiting SteamVR make sure to exit from within Stream, e.g. press the Left Touch menu button and exit Steam using the exit icon in the lower right corner.

  Note: The Rift needs to be set as default microphone device for Oculus Tray Tool to hear you. This needs to be done before voice control is initialized, which is when Oculus Home launches. Use the "Set 'Rift' as Default" option or manually make the change before Oculus Home starts.

- Activating Voice Commands
  On the Activation tab you can select how to enable Voice Commands. Each time Oculus Tray Tool starts listening, a sound will be played to indicate that it's listening, and a slightly different sound will be played when it stops listening. You can disable these audio feedbacks by ticking

the “Disable audio feedback” checkbox. The title bar of Oculus Tray Tool will also change to indicate it is listening.

The following options are currently available for activating voice commands:

- Voice activated, continuous
  Selection this option requires you to speak the phrase set for “Enable Voice Control” once and then all other commands become available. To stop listening, speak the phrase set up for “Disable Voice Control”.

- Voice activated, repeated
  This option requires you to speak the phrase set for “Enable Voice Control” each time before you want to issue a command. After each command, Oculus Tray Tool will stop listening. Here, you could for fun add a name to the list of phrases for “Enable Voice Control”, like for instance “Allie”. To then show for example the current pixel density, you would the speak “Allie, **enable confirmation sound plays** show pixel density **disable confirmation sound plays**.

- Keyboard activated, continous
  Lets you set up a key on your keyboard to use for activation voice control. Press it once to enable, and again to disable. Select “Keyboard” in the “Select device for button/key activation” drop-down list. The click the crossed-out ear icon to start capture, and press the key you wish to use.

- Keyboard activated, Push-To-Talk
  With this option, you need to keep the selected key pressed down while issuing voice commands. Let it go to stop listening.

- Joystick activated, continous
  This option let’s you choose a button on a joystick to use for voice commands. Same principle as for Keyboard activated, continous.

- Joystick activated, Push-To-Talk
  Same principle as Keyboard activated, Push-To-Talk but using a button on your joystick instead.

- Editing Voice commands
  Click the Edit button next to the Voice commands dropdown list to get a list of the currently available voice commands. Here you can also edit the list and add your own phrases, and have multiple phrases for the same function. Oculus Tray Tool currently uses the Microsoft.Speech API which requires no user voice training. It currently best understands English.

- To edit a voice command, simply click on the row and the properties for it will open up. For the System Commands you can only edit the spoken phrases. When any of the phrases is spoken, the selected Action will be performed.

All of the phrases are spoken as they are written, with the exception of the Set Pixel Density function. Here you also speak a value when you speak the phrase. The value corresponds to the Super Sampling value you want to apply, and these values are hardcoded and cannot be edited. If you for example want to set super sampling to 1.5, and the phrase for this is "*set super sampling*", you would say "*set super sampling one point five*". Valid range of values are 0, 1.1 to 2.5 in 0.1 increments. If the application understood you it will apply and confirm the action.

If you are currently inside a VR game/application, you will need to restart the game for a new Super Sampling value to apply. If you are currently in a game that has a profile, the profile will be updated with the new value. This also applies to ASW.

An example of using voice commands is when you have a game that you do not yet have a profile for, and you want to find the best Pixel Density/Supersampling value for it.
Put your Rift on but before starting the game, set a Pixel Density value by speaking the phrase for that action. Start the game, and then use the "Show Pixel Density" voice command to bring up the overlay that displays the current Pixel Density setting to confirm what setting you have. Play the game and see how it runs, use the "Show Performance" voice command to check how it runs and if there's any overhead, meaning you might be able to push Pixel Density just a little further still. Not satisfied with the current setting? Quit the game, set a new Pixel Density value using the voice command and then start the game again. This way you can use Voice Commands to fin the best setting for that game, all without taking the Rift off. When you are satisfied, simply go back to OTT and create a Profile with the settings for that game.

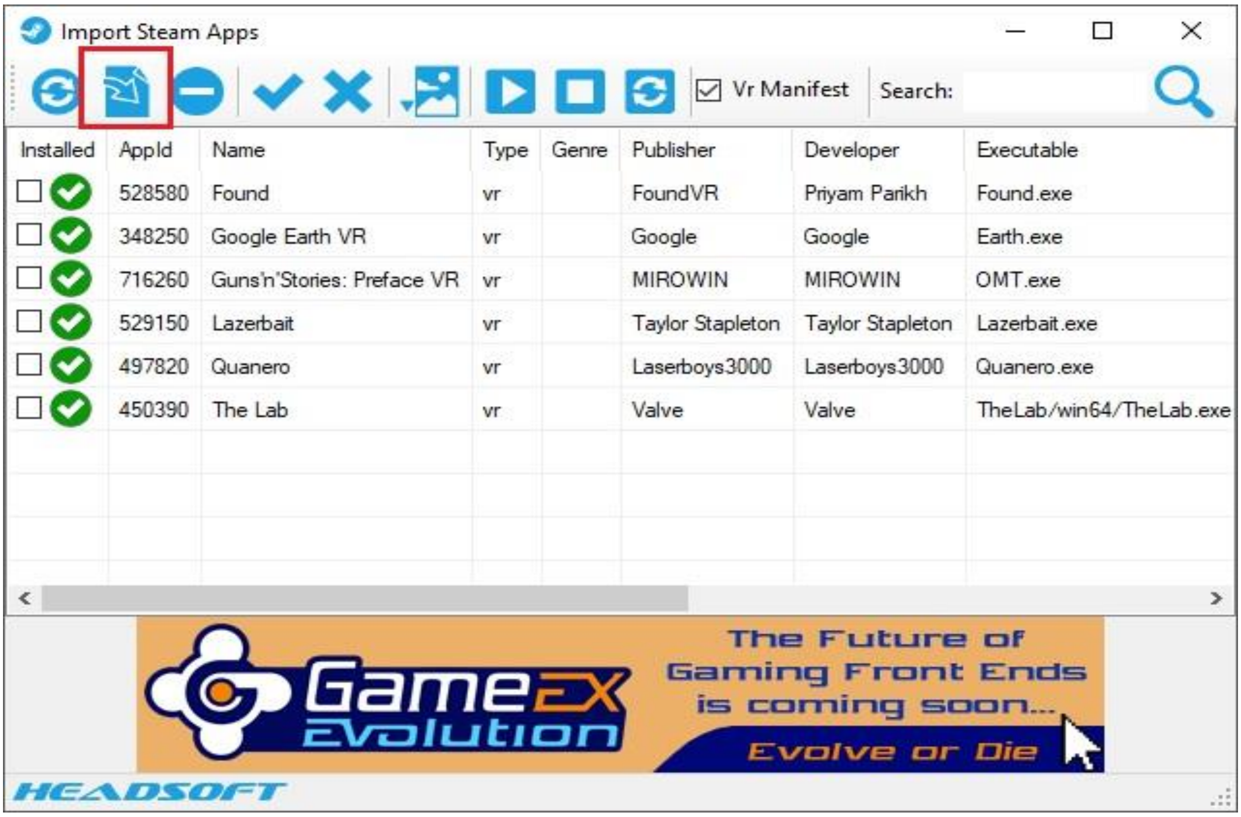
- Confidence
  The confidence value tells OTT how sure it must be that you spoke a certain phrase before performing the action of that command. This higher this is set, the less likely it is for OTT to misunderstand you. But it also puts a higher demand on the quality of the spoken words, e.g. how well the user speaks English and how good the microphone being used is. If English is your natural language then you can probably put this slider quite high, but if your English is not the best, then you might to set this to a lower value. You will need to experiment a bit to find the setting that works best for you.

- Game Library: Click to view your Library and perform actions such as replacing icons for third-party apps or view the Properties of your apps. Third-Party applications properties can also be edited, and you can change things like launch options, change display name of the apps and even change what .exe should be called on launch.

  - Title colors. The color of the title text in the library depends on if the game has a profile or not. Red title text means it does not have a profile, Green means it has a profile and Blue means it probably has a profile, but the games launchfile differs from the profiles launchfile. This is normal if you have changed what .exe OTT should monitor for this game.

- You can launch games directly from the Library. This should ensure that the profile is applied. Try this if you are having issues with getting the profile for a specific game to apply.

- Steam Library: This feature implemented by Oculus forums user @headkaze allows you to bulk import you Steam games without the need for starting each one first. Clicking the View & Import button will open a new window listing all your Steam games.
  By default, only those that are VR compatible are listed. To view all, uncheck the "Vr Manifest" checkbox. Select the Steam titles that are not already in the library, indicated by a red X. Click the Import button, as highlighted below, to import them and make the visible in Oculus Home. If you want to remove a title, simply select it and click the Remove (-) button.



You can also start you Steam games from this windows, which will also add then to your Oculus Home view. This feature is currently in Beta and might need a bit of tweaking. Please report any issues to ApollyonVR@gmail.com or send me a private message on the Oculus forum, @ApollyonVR. And make sure to visit @headkaze over at http://www.headsoft.com.au/ and check out all his other great stuff!!

- Visual HUD Overlay:
  Changing this while in Oculus Home or a game will bring up an overlay window with some useful information regarding current status and performance. Use these to tweak out the best settings for each game.
  https://developer3.oculus.com/documentation/pcsdk/latest/concepts/dg-hud has proper explanations on all of these, brief overview below.

  None: Close any open overlays.

  Pixel Density: Show overlay with Pixel Density at the bottom.

  Performance: Show Performance overlay. Use this to find the best Pixel Density for your games. The left graph shows FPS, you want this to be green and around 90-88 FPS for optimal experience. Experiment with different Pixel Density settings and observe how the graph changes. Remember, restart the running VR game/application between changing to Pixel Density values.

  ASW status: Show the ASW status overlay. Here you see if ASW is Off or Auto.
  You can have this overlay open, then use either voice commands or the ASW dropdown to see the status change in the overlay.

  Apart from the above, there are also some more in-depth overlays if you really wanna go swim at the deep end and check everything.

Service & Startup

These options are fairly self-explanatory so I'm not going to go through them all.
Some people like to keep the Oculus service stopped when not using the Rift, so the options regarding the service start/stop are mainly for them.

Note: If you have issues with Global settings like ASW and Supersampling not applying properly when you use OTT to manage the Oculus Service, it might be due to timing issues.
If so, you can edit the user.config file (the location of this is printed in the ott.log on each start)

1) Open the user.config file and find the parameter named SleepAfterServiceStart
   If it is not there simply create it like below

   ```
   <setting name="SleepAfterServiceStart" serializeAs="String">
       <value>3000</value>
   </setting>
   ```

2) The default value is 2000ms, so change this to a higher value, like 3000.

Note: Oculus Tray Tool will not be able to start the service if it is set to Disabled. Set it to Manual or Automatic if you want to be able to control it using this application.

- Spoof CPU ID on tool start: Check this box to change the CPU ID when the tool starts. Use this option if you see the message "*Your computer doesn't meet the recommend requirements*", and want to get rid of it. This changes the "ProcessorNameString" value in the registry, making the Oculus service believe you are running an i7-4770K @ 3.90GHz or an Ryzen 7 1700X if you're running with an AMD CPU. Your old CPU ID is always restored when Windows boots up. It is also saved in Oculus Tray Tool settings, as well as written to the ott.log file. Unchecking the box will also restore the old CPU ID. The Oculus service needs to be restarted for this to work, so having it checked when the tool starts will force a service restart.

- Send Oculus Home to tray: Setting either of these options will minimize and basically hide Oculus Home. To restore it, right-click the Oculus Tray Tool icon in the system tray and click "Show Oculus Home", or close Oculus Tray Tool.

  If you are having issues with Home not always minimizing to the system try you can try and edit the timeout.

  1) Open the user.config file (the location of this is printed in the ott.log on each start).
  2) Find the parameter named SleepAfterHomeStart. If it is not there, simply create it like below

     ```
     <setting name="SleepAfterHomeStart" serializeAs="String">
         <value>3000</value>
     </setting>
     ```

  3) The default value is 2000ms, so try setting this to an higher value like 3000.

Log tab

The Log window is displayed by going to the Log tab. The text of this tab will differ depending on how the startup went.

- *Log*: All is good. Go play some games!
- *Log (Warning)*: A warning has been reported. The log window should tell you what the issue is. If it is reporting that a device had power management enabled, simply right-click the message to disable PM on all reported devices.
- *Log (Error)*: Something has gone terribly wrong! Take a screenshot of the log window, grab the ott.log and send that to apollyonvr@gmail.com or post on the Forum over at www.apollyonvr.com.

Quest Link

Here you can set some performance options if you have a Quest and Link cable. There are some presets to choose from but you can also edit the settings as you like. These settings have no effect If you do not use an Oculus Quest or Quest 2.

- Permanent AirLink
  This option uses code written by GitHub contributor pd29 (https://github.com/pd29/oculus-airlink-enabler). It patches Oculus so that the AirLink feature is not disabled after 12 hours. This is a multi-step process:

  1) First the Chocolatey Packet Manager is installed
  2) Next, Node.js is installed by Chocolatey
  3) Then, Asar is installed by the Node.js Packet Manager
  4) OTT makes a backup of <install_path>\Support\oculus-client\resources\app.asar
  5) Asar is then used to extract the app.asar package and inject pd29's custom code

6) Asar repacks the new app.asar package

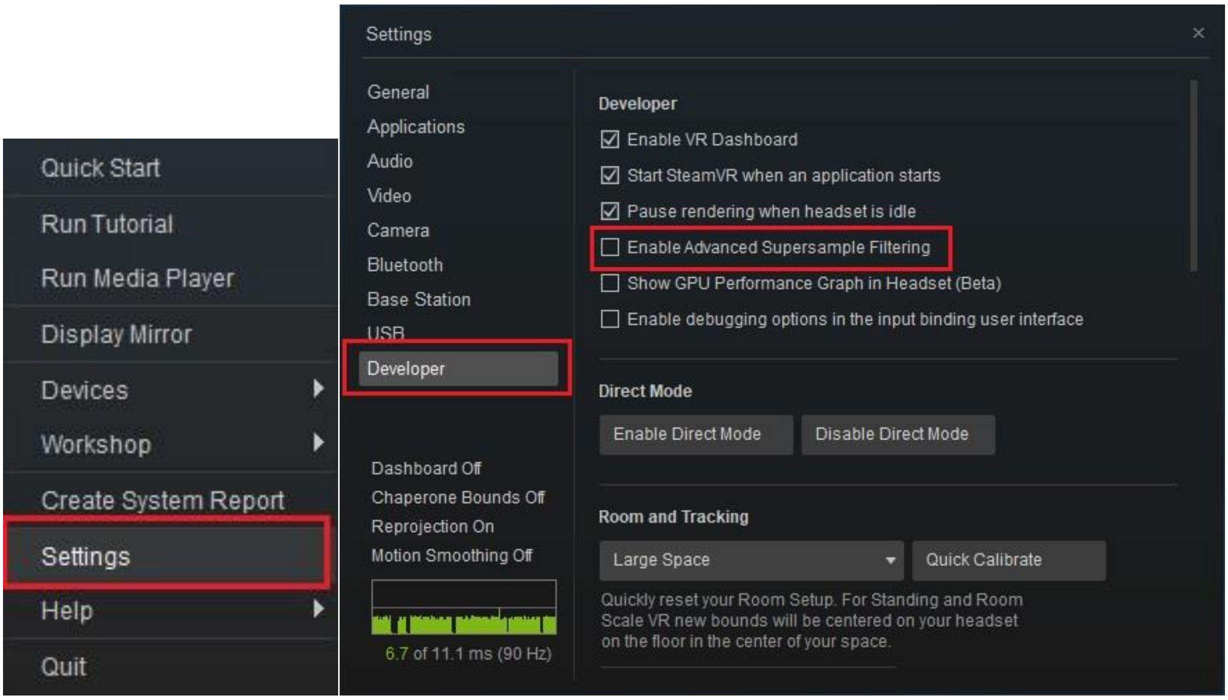
Advanced Tab

- Oculus Debug Tool: Use local debug Tool
  Should remain unchecked. By default Oculus Tray Tool will use the debug tool that is shipped with your Oculus installation. If for some rare reason you want to use another version of the debug tool, copy that version into the OTT installation folder, and then tock this box.

- AppWatcher: Start on OTT start
  Should remain unchecked. The AppWatcher is an OTT internal process that is responsible for detecting game and app starts, so that OTT can apply the profile settings. This internal process is automatically started when Oculus Home is detected as running. Back in the Good 'Ol days, one could run a VR app without having Home running, and as such this box needed to be ticked. It should not be used unless you are doing some experimenting.

Steam VR Settings

If you experience mixed results when setting ASW and Super Sampling for Steam games, try disabling Advanced Supersample Filtering in the Steam VR settings. To do this, start Steam, then start Steam VR by clicking the VR icon in the top right corner. When it has started up you access the settings by right-clicking the VR icon in the Windows system tray (where the clock is, click the little ^ symbol to access all apps running in the tray). Once you are in settings, click Developer and then untick the option for "Enable Advanced Supersample Filtering".



Ignored Apps

Oculus creates JSON manifest files (basically text files) for every app that is started when Home is running. Even non-VR apps like Firefox or Calculator gets added. OTT find games by reading these files, and using several different methods to try and determine if they are VR apps or not.
If OTT determines the app is probably not a VR app it gets added to an internal “Ignore list”. Due to Oculus being a constantly changing entity, it can be that OTT makes a mistake in filtering out these apps. You can view the Ignore list and manually include apps that you want in the OTT Library. To do this, go to “Options -> Show Ignored Apps” in the Library, and you will see a list of apps that OTT is ignoring. Check one or more apps and click “Include Selected”. This will add the app to the Library, and also make it available for selection in the “Create/Edit Profile” dialog.

Super Sampling/Pixel Density recommendations

What is Super Sampling anyway? Super Sampling Anti-Aliasing (SSAA) is a technique for anti-aliasing, e.g. make the image look sharper and better. It works by rendering frames at a higher resolution than the display resolution, then squeezing them back down to size. Check out http://www.dahlsys.com/misc/antialias/ for an overview of different anti-aliasing techniques.

So what is the best Super Sampling value for game X? Pretty hard to tell as that depends on the power of your computer. Super Sampling is very taxing on your hardware, and you need a pretty beefy machine to use high settings with ASW off. ASW is pretty awesome and I myself have it enabled, allowing me to run high SS without many issues. A good way would be to use the different Visual HUD overlays to get information on performance to tweak things to your liking.

Asynchronous Spacewarp

Commonly abbreviated as ASW, Asynchronous Spacewarp is a tech by Oculus that increases the perceived framerate of a VR application. Basically it forces the application to render at 45 FPS, then estimates the position of your hands and the scene itself, adding "synthetic" frames between each rendered frame to free up resources for positional calculations. It does this so well it makes you perceive the application as running at 90 FPS, usually without seeing many incorrect frames.

Check out https://developer.oculus.com/blog/asynchronous-spacewarp/ for a more in-detail explanation of ASW.

Page file considerations

I recommend having a page file as this seems to increase stability at least when running VR games. By default Windows manages the page file, but I've found that manually setting Minimum and Maximum values to match that of your graphics card memory size has helped with game crashes. Might just be me, but in either case, below are the details for how to manage the Windows page file.

1) Use an administrator account to log on to Windows 10.
2) From the desktop screen, right-click the Start button to open its context menu.
3) Click System.
4) From the left pane of the System window, click Advanced system settings.
5) On the System Properties box, ensure that you are on the Advanced tab.
6) Click the Settings button from under the Performance section.
7) On the Performance Options box, go to the Advanced tab.
8) Click the Change button from under the Virtual memory section.
9) On the Virtual Memory box, uncheck the Automatically manage paging file size for all drives checkbox.
10) From the available list, click to select the drive on which Windows 10 is installed. (C:)
11) From below the list, click to select the Custom size radio button.
12) In the now-enabled fields, type the minimum and maximum size of the Pagefile in megabytes (MB). I would recommend setting the Minimum size to at least the same as the amount of memory your graphics card has. For Maximum size increase this value by 2GB.
13) Click Set and then click OK.
14) Restart your computer when you're done.

Overclocking your graphics card

Overclocking can be a great way of increasing the power of your PC, but when it comes to graphics card it may lead to some very undesirable results as the card will be under a lot more stress. If you are running your GPU overclocked and experience hangs or other weird issues, especially when running VR games at increased Pixel Density, the first thing to do is to lower or possibly even disable the overclock on the GPU to see if that resolves the issue.

Troubleshooting

- The application keeps getting deleted
  Some antivirus applications, in particular Avast, seems to dislike OTT and flags it as a IDP.Generic malware. This is what is called a "false positive" where the AV application detects "unusual activity" when OTT reads the registry. Make an exclusion for the file OculusTrayTool.exe in Avast and all should work fine.

- Sound stops when I enable Voice Commands and/or the Audio Switcher
  To resolve this, decrease the sampling rate on the Rift to 44100hz.

  1) Right-click the Volume icon in the Windows system tray, down by the clock.
  2) Select "Playback devices"
  3) Select "Headphones (Rift Audio)"
  4) Click "Properties"
  5) Go to the "Advanced" tab
  6) Select "44100hz (CD Quality) in the dropdown.

- Voice commands are not working

  1) Voice recognition, if set to Enabled, is initialized when Oculus Home launched.
     At that time the voice engine will start listening on whatever microphone is set as default. Make sure that either the Rift, or another microphone you intend to use, is set as default *before* Oculus Home is launched. You can check this by right-clicking the sound icon in the system tray and selecting Recording Devices. Verify that the device you intend to use is set as default, and that it is registering you voice by saying something. If the sound bar is not moving the device is not receiving any input. Oculus Tray Tool has options for automatically setting the Rift as default audio or microphone device to make this a bit easier.

- Not all apps are visible
  See "Ignored Apps"

- Clearing all settings to default
  If you are having issues with something acting weird, you can try and reset a few settings to default. This is done by either going to the Advanced tab and click "reset all settings to default" or by first closing Oculus Tray Tool and then deleting the following folder: C:\Users\<YourUsername>\AppData\Local\ApollyonVR.

  You can also clear settings by running the application from an elevated command-prompt using "OculusTrayTool.exe –r"

- Run in debug mode
  To get more info in the ott.log you can run in debug mode. You do this by easiest by going to the Advanced tab and clicking "Restart in Debug Mode". The application should shut down and restart. If it does not restart by itself, simply start it again and it will be running in debug mode. The ott.log file located in the installation folder will now contain a lot more info. If you have an issue or suspect a bug, run in debug mode and send me the ott.log.

  You can also enter debug mode by running the application from an elevated command-prompt using "OculusTrayTool.exe -d"

- Force settings update
  When Oculus Tray Tool is updated to a new version, the settings from the previous version is migrated over to the new. In case you run a beta version the version number might not be updated when I add new settings. You can force an update by running the application from an elevated command-prompt using "OculusTrayTool.exe -u". This is recommended for all beta testers to do on first start with the Public release. This also happens when you run consistency check from the Advanced tab

- Run Consistency check
  From the Advanced tab you can click "Run Consistency check" to check that all profile paths in the ott.db are correct and exists. Running a Consistency check also makes sure the user settings file is migrated and clears all apps from the "ignore list" in ott.db.